Fordism, Social Reproduction and the Global Division of Labor

Abstract: This article challenges the view of Fordism as a development local to American industry. The dynamics that have been analyzed under the heading of Fordism were neither local to the Midwest, nor limited to the productive and reproductive characteristics in the US and Europe. Rather they were, from the very start, global dynamics. This is especially clear in the connection between the automotive industry and the rubber industry in the Global South. The phenomena commonly identified with Fordism such as high pay and an impetus to working class consumption cannot be separated from the extreme exploitation in the rubber producing regions of the world that made them possible. At the heart of Fordism is the creation and maintenance of a stratified labor force, manipulated by differentiated but interlocking relations of social reproduction. The reproductive regimes in the US and the tropics should not be analyzed as separate phenomena, but as two constituent elements of the Fordist dynamic of production. Drawing on literature from three continents, this paper argues for the relevance of a global perspective in the study of the early automotive industry.

Fordism, Social Reproduction and the Global Division of Labor

Fordism has been a slippery concept within historical writing over the last 100 years. Although labor historians and economists agree that it is a central moment in both American and global economic history, there is little agreement on what Fordism actually *is*. In 1990, in a talk humorously titled "What in the F—'s name is Fordism", British sociologist Simon Clarke noted the imprecision in discussions of Fordism and whatever non-Fordist capitalist epoch is supposed to be its successor: "Much energy has been expended in debating the diffuse characterizations of 'Not-Fordism'. However, much less attention has been paid to the characterization of Fordism" (Clarke 1990: 13). Even the popular legacy of Henry Ford himself is deeply ambivalent. He was a benefactor who improved the lives of his workers and a tyrant whose Sociological Department spied on them in their homes; he created the modern middle class but also initiated the culture of consumption that is currently eating the planet; he was a humanitarian who founded schools for his mostly immigrant employees and he was a vociferous anti-Semite. Only one thing is uniformly agreed upon: Fordism represented a genuine break with existing patterns of production in capitalist society; it was a revolution.

The boundaries of this revolution are blurred. Attempts to define it find themselves faced with irreducible contradictions. Daniel Watson identifies two main approaches to Fordism: a narrow view of Fordism as a system of mass production centered around Henry Ford's innovative assembly line, and a broader view of Fordism as a "phase of capitalist development taking place several decades after the heyday of the Ford Motor Company" that is characterized by a "careful balance in the American economy between mass production and mass consumption and an increase in the wage rate and social provisions for workers" (Watson 2019: 2). In this

broader view, the Fordist era was initiated by Henry Ford's productive innovations, but was not fully realized until decades later when his policies were generalized to the rest of the industrialized world.

Each of these definitions is unsatisfactory. In the first place, "[t]here was nothing original in either the detail or the general principles which Our Ford applied to the production of motor vehicles" (Clarke 1992: 16), and so it makes little sense to isolate them from their social context. But if Fordism is to be defined by the unique articulation between these general production principles and social policy, then why have analysts found it "difficult to discover any examples of 'Fordism' in its pure form, even in the United States" (Clarke 1988: 62)? Inflation quickly caught up to the Five Dollar Day, and the Sociological Department was dissolved in 1920-21, effectively ending Fordist paternalism and morality bonuses (Meyer 1981: 197). In the US, the generalization of social provision to workers was not effected by private capital but by the state, with Franklin D. Roosevelt's New Deal. This flew in the face of both Henry Ford's ideology and his policies, as he was a fierce opponent of the welfare state (Fine 1958), and had placed restrictions on his worker's access to social benefits.

The primary legacy of Fordism is rather one of stratification. Simple narratives about the Fordist worker and the proliferation of the family wage do not hold up to scrutiny. In Michigan's auto factories in the first decades of the twentieth century, local workers were increasingly divided into two camps. On the one hand, workers increasingly performed manufacturing tasks that were decomposed into minimal parts, rationalized to the extreme and voided of intellectual stimulation. On the other hand, a great class of managers and engineers was cultivated to design the machines and processes that would combine the decomposed labor of others into finished

products - following a managerial revolution that Alfred D. Chandler called the ‘visible hand’ of management (Chandler, 1977). Outside the factories, the living conditions of workers were similarly divided. Skilled and semi-skilled workers, mostly white, were transformed into large-scale commodity consumers and inducted into a class of homeowners, while racialized and menial laborers often lived in crowded slums and still relied significantly on informal labor for survival.

Of course, the labor that contributed to the manufacturing of automobiles was not confined to factory work. The operations of the Ford Motor Company were embedded in a global network of production. Raw materials had to be imported - especially rubber which could only be sourced from the Global South. The stratified labor market in industrial Michigan was mirrored by a global division of labor, in which wild rubber harvesters in the Amazon and the Congo, and to a lesser extent the later rubber plantation workers in southeast Asia, occupied an intensely degraded position. Wage labor was not the dominant labor relation in the rubber regions of the world, and when Northern capital intervened in the rubber industry, it created quite different modes of exploitation. These modes of exploitation were accompanied by different patterns of consumption as well. The rubber workers were made to be dependent on the market for the means of survival, and especially on the importation of canned goods and preserved food from the Global North. But they were not consumers of domestic commodities, sheltered by a commodity house and supported by de-commodified reproductive labor. Their precarity was exploited by capital to effect an extension of the market into the private lives of producers on very different terms than in the Global North.

This article takes up early labor stratification in the automotive industry and the methods used to create and discipline it. In order to analyse the connection between conditions in the core and periphery, this articles makes reference to the commodity chain. Wallerstein and Hopkins used the term commodity chain to conceptualize the interconnectedness of global capitalism: "take an ultimate consumable item and trace back the set of inputs that culminated in this item - the prior transformations, the raw materials, the transportation mechanisms, the labor input into each of the material processes, the food inputs into the labor" (Hopkins and Wallerstein 1977: 128). The commodity chain is the axis along which Fordist firms tried to integrate, but this chain includes more than just raw materials and the labor that works them: it connects to "the reproduction of the labor forces involved in these productive activities" (128). Along the commodity chain, different materials enter the production process through a variety of labor regimes, from wage labor in the factories to debt-bondage and enslavement in the rubber fields.

The axial division of labor has as a corollary an axial division of social reproduction, by which lower-waged laborers and high-waged laborers are supported by different household structures. This article makes the case that the social reproductive dynamics in the rubber producing regions of the Global South are not externally articulated with Fordist reproduction, but internal to it. Automotive production required several different types of material and labor inputs, and these corresponded to different regimes of social reproduction. In particular, they took advantage of a different articulation between the rationalization of labor, and the expansion of commodity consumption among the workers. The scope of this article will focus on the origins of Fordism: the production of automobiles in the first decades of the twentieth century.

The Contradictions of Fordism

In the popular imagination, Fordism comprises two moments of personal genius: Henry Ford's implementation of the automated assembly line and his institution of a family wage with the Five Dollar Day. Faced with the mutually exacerbating problems of rising demand and increasing employee turnover, Ford took the innovative approach of intensifying and de-skilling labor while at the same time increasing its pay (Dassbach 1991). These principles were combined in Ford's notorious River Rouge factory, a single complex that took raw materials as inputs turned them into cars. In the meantime, the laborers who worked in this factory were middle class workers who could afford to purchase the commodities they were producing. At one and the same time the efficient production methods transformed automobiles from a luxury good to a near-necessity, and transformed the workers from struggling laborers into participants in the bounty of production.

This popular conception is problematic on several counts. Ford's production practices were enormously successful, but the core idea of a powered assembly line was borrowed from Chicago slaughterhouses and had first been employed in the automotive industry by one the EMF company which "installed a mechanized assembly line to boost chassis output to 100 per day" (Lewchuck 1989: 20). Ford would not install his first line (also for chassis production) for another two years, in 1913. Even after Ford's unprecedented success with rationalizing production, his methods would often be taken and improved upon by competitors - most notably General Motors. Ford's real innovation was the extreme degree to which he took the rationalization of production, and its combination with an aggressive social policy. His famous Five Dollar Day was not a universal wage increase, but a profit share given only to employees

whose private lives were vetted by agents of the Ford Sociological Association. In addition to investigations into the stability of the workers' families, "the company wanted to know whether or not the workman owned or was purchasing property or a home, whether or not he had a savings account in a bank, and whether or not he was in debt" (Meyer 1981: 131). When combined with wages that exceeded those offered by competitors, this meant that Ford's prized workers were rooted in the community and thus less able to seek out employment elsewhere. The question was not one of boosting the freedom of workers by increasing their pay, but of limiting their freedom and compelling them to stay in their jobs.

Scholarly attempts to resolve the contradictions of Fordism have usually involved a sort of analytical surgery, either removing the social-reproductive aspect and focusing on Henry Ford's specific management practices, or excising Ford almost entirely and treating 'Fordism' as synecdoche for larger structural changes in the global accumulation of capital in the mid twentieth century. Each of these approaches is untenable. The first reduces Fordism to a loose collection of stolen ideas that were implemented with a ruthless intensity, but which explain very little about the history of capital. The second attempts to establish an ideal type of capitalist society that bears little resemblance to any actually existing world. It inevitably devolves into a list of precursors, exceptions, and modifications that overwhelm the type itself: Taylorism, Sloanism, Toyotism, Keynesianism, Post-Fordism, Neo-Fordism, etc. Sociologist Simon Clarke criticizes this tendency, saying "to explain these variants as failed versions of one ideal-type implies that the latter is in some way more real or more fundamental than the former" (Clarke

1990).[1] Indeed, it is hard to see why Fordism would stand out in this list as the determining element.

The irresolvability of these contradictions are made clear in the work of the regulation school, a group of French scholars who devoted considerable time to defining and delimiting the concept of Fordism. Their scholarship was rigorous, but flawed. Within this group, Michel Aglietta provides the classic structuralist account of Fordism as a generalized set of production principles, historicized within the long cycles of historical capitalism. For Aglietta, although Fordist practices may have originated as specific policies within the auto industry, Fordism represented a societal-level shift in the relations of production and the reproduction of capital. In *A Theory of Capitalist Regulation*, he criticizes the dominant economic theories of capitalist equilibrium for being, among other things, ahistorical and abstract (Aglietta 1979). Against this, Aglietta denies "that what exists does so automatically", necessitating a theory of social reproduction, and he claims that "when actual social systems are studied, historical experience confirms that *transformation means rupture, qualitative change*" (12). In other words, although Henry Ford's specific policies may have been solutions to the immediate problems in his own factories, their eventual widespread adoption does not indicate that they tapped into some transhistorical and abstract need for workers to receive higher wages. Rather, it indicates that these policies addressed a *historical* condition that may not have existed before, and may change again in time.

Taking the US economic as his object, Aglietta elaborates a theory of capitalist regulation as an attempt to "formulate in general laws the way in which the determinant structure of a

[1]This phrase does not appear in the version of this paper published in *Fordism and Flexibility: Divisions and Change*, but in an earlier (unpaginated) draft presented at the 1990 British Sociological Association Conference available at this web location: https://scispace.com/pdf/what-in-the-f-s-name-is-fordism-2bwbsf1e0m.pdf

society is reproduced" (13). He positions the economic collapse of the 1930s as a result of overaccumulation and underconsumption. Specifically, the average worker was paid so little in the late 19th and early 20th centuries that they had to support themselves through a significant amount of informal labor, and therefore could not contribute to the valorization cycle by purchasing and consuming commodities. As capital entered a crisis in the 1930s, producers sought out new ways to maintain profitability. This would mean restructuring society to expand the domestic market and increase the class of consumers. As producers cast about for a model, Fordism, at this point a decade and a half old, stood out as a generalizable answer to changing economic conditions.

Fordism figures into Aglietta's account as an 'intensive regime' which "creates a new mode of life for the wage-earning class by establishing a logic that operates on the totality of time and space occupied or traversed by its individuals in daily life" (Aglietta 1979: 71) Fordism was characterized by an unprecedented expansion of capitalist interest and control over the lives of its waged workers beyond the shop floor. In order to stave off another crisis of overproduction and underconsumption, the daily lives of workers would have to be rebuilt on the foundation of commodity consumption. Aglietta terms this reorganization of social life a 'mode of consumption' which is oriented towards the renewal of the energy expended in labor processes and the maintenance of a continual supply of labor power.

As an articulation between production and consumption, Fordism comprised several specific innovations linked to more general social reconfigurations. On the production side, Aglietta identifies the automated assembly line as Fordism's "characteristic labour process" (Aglietta 1979: 117). The assembly line separated mental and manual labor to a previously

unseen degree. Instead of managing the relationship between several related tasks, workers performed simplified rote tasks over and over: “the labour process [was] converted from a dense network of relationships between jobs, with intermediate products passing back and forth … into a straightforward linear flow of the material under transformation” (118). In Ford's factories, the production of a car was decomposed into 7,882 separate tasks, each of which was carefully documented and classified. The workers were similarly rationalized - not humans, but various assemblages of limbs: "The lightest jobs were again classified to discover how many of them required the use of full faculties, and we found that 670 could be filled by legless men, 2,637 by one-legged men, 2 by armless men, 715 by one-armed men, and 10 by blind men" (Ford and Crowther 1922: 108).

The mental component of labor was excised from the ambit of the general laborer and given to a growing class of managers and process engineers. Workers became fixed to clearly and rigidly defined roles that were “determined by the configuration of the machine system” which was “planned in engineering departments which were themselves subject to the division of labor” (Aglietta 1979: 118-9). In this sense, the Fordist revolution went hand in hand with the managerial revolution described by Alfred Chandler. As the scale and scope of production increased, and as jobs were broken up into smaller and smaller elements of an intricately designed whole, the "division between the labour of head and hand" (Sohn-Rethel 1978) was deepened. Thought was no longer a part of the act of labor itself, but something that had already been completed in the past and invested into the creation of a proscribed list of infinitely repeatable movements. The Fordist workplace began to look more and more like that described

by Marx: living labor ruled by the dead labor that was embodied in the planning and manufacturing of the machine system.

This process of rationalization extended beyond the shop floor. Along with the revolution in production, Fordism reconfigured the social and private lives of the workforce. Aglietta notes that "for the first time in history, Fordism created a norm of working-class consumption in which the general ownership of commodities governed the concrete practices of consumption" (Aglietta 1979: 158). Two commodities stand out as governing this norm: the automobile as a means of private transportation, and standardized housing. The adoption of these two major commodities transformed other patterns of household production. Households began to substitute the purchase of ready-made products like canned food and clothing for space-intensive home production, such as butchering and tailoring. Furthermore, the nineteenth century pattern of roaming peddlers bringing such necessities as pots, pans or medicines, directly to the tenement buildings where workers lived, was replaced by workers taking their now-indispensable cars on trips to the grocer and hardware store (Cowen 1983: 71-85). All of these commodities transformed the urban spaces in which the working classes were renewed, and their relatively high cost "presupposed a vast socialization of finance, and correlatively a very strict control over workers' resources and expenditures" (Aglietta 1979: 159). However, Aglietta notes that the rise of certain family norms within a consuming class of workers was never quite general: Fordism also deepened the stratification of the working classes, which "forms part of the concrete conditions of reproducing the wage relation" (137).

Aglietta's account of Fordism as an historical articulation of means of production and consumption responding to the global needs of capital accumulation is a compelling one, but it

fails in its (non)treatment of imperialism, and its abstraction of US capitalism from the system of global capitalism. In the introduction to *A Theory of Capitalist Regulation*, Aglietta defines imperialism as a system "through which one state manages to influence a series of other states to adopt a set of rules that are favorable to the stability of a vast space of multilateral commodity relations guaranteeing the circulation of capital" (Aglietta 1979: 32). After posing this anemic definition, he pushes it aside, claiming that "there could be little question of analyzing in the present work the system of contemporary states and the transformation that the crises of the 20th century have produced in inter-state relations" (32) But he is unable to escape the weakness of his conception of capitalist hegemony - it undermines his entire project. In his attempt to analytically isolate America from the Global South, he misapprehends the extent to which American industry both exacerbated and was dependent on an unequal global division of labor. The intensification of exploitation across state lines undergirds capitalist accumulation in Global North states. His insufficient definition of imperialism as management inflicted by one state on several with the goal of broadening the circulation of capital prevents Aglietta from appreciating the global scope of capitalist regulation and reproduction.

In "Towards Global Fordism?" Alain Lipietz, another scholar from the regulation school, attempts to more adequately theorize the connection between the Fordist core and the periphery. Lipietz questions whether we can even "speak of a world-wide extension of Fordism," noting that during the late 1960s and early 1970s, the economic periphery experienced uninterrupted growth while the economic core went into a production crisis. Indeed, the industrialization of the periphery was not a slow march following in the footsteps of the core, but was its own specific development, conditioned by and conditioning the dynamics of accumulation in the United

States and Europe. By the time 'Fordism' reached the Third World, it had devolved into another one of its non-ideal, mutated forms. Lipietz describes a Global South Fordism stripped of its impetus to consumption as "sub-Fordism" and "'Bloody' Taylorization" (Lipietz 1982).

Although Lipietz's work is rigorous and empirically rich, it is undergirded by a flawed methodology. He maintains Aglietta's idealist assumption that the Fordism of the US can be analytically extracted from its own contemporary global context. He then looks at the structures of accumulation in the Global South decades later and compares them to this ideal type. The implicit question is not 'how did the structure of accumulation in the South develop?' but 'why did Fordism fail to be exported as-is?' It sidesteps the question of how production in the South contributed to the development of Fordism in the core.

A more appropriate approach to the global scope of Fordism is to trace the specific ways in which the automotive industry created new relations of production and consumption in peripheral countries, and how these relations contributed to the accumulation of capital in the core countries. The automotive commodity chain - the global network of extraction and production processes that transform raw materials into finished goods - extended into the tropical regions where rubber grew. As an essential material unavailable in the US and Europe, rubber and its conditions of production expose the fundamentally global nature of the production of automobiles. Cars could not be produced if rubber was not available - and their production would not be profitable if it was too expensive. In the drive to procure sufficient quantities of cheap rubber, European and American capital pulled the rubber producing regions in the Amazon, Africa and Southeast Asia into their sphere of influence, but on deeply unequal terms.

Several factors contributed to the underdevelopment of the rubber regions: the demands of Northern industry for cheap rubber, Colonial governments with no interest in the development of stable and autonomous economies in the Global South, and unscrupulous local actors happy to exploit their countrymen for a marginal share in the profits of capital. Each of these factors interacted with the same categories that were so important to Henry Ford: the possibilities of rationalizing processes of production and the substitution of commodity consumption for subsistence labor. These two elements of re/production were manipulated in different ways than in the US Midwest, but they were unified as part of a "logic that operates on the totality of time and space occupied or traversed by its individuals in daily life" (Aglietta 1979: 71). To sideline the exploitation of rubber workers, and the exclusion of Global South industry from the profits of the automotive industry is to misrecognize Fordism as a category of the capitalist world-system.

If we have questioned the definition of Fordism as a systematic regime of accumulation, and its validity as a mode of regulation divorced from both state action and the global contours of capital, we might reasonably ask what use the concept of Fordism provides, and what value we can gain from the regulation school's treatment of it. Two of Aglietta's core interventions - Fordism as a specific qualitative change in the articulation of production and consumption, and the increased stratification of the wage-earning class - are crucial, but they must be reformulated as questions: what concrete factors led to the particular relationship between production and consumption in the early automotive industry? Furthermore, we cannot define Fordism so narrowly as to exclude the conditions of production in the Global South which made automotive production possible. The production of rubber in the Amazon, Africa and Southeast Asia was not external to the automotive industry, but internal to it. With this in mind, following sections will

investigate the articulation between production and consumption in the rubber industry, and the stratification of the rubber workers, with special attention to the development of commodity consumption and attempts at the rationalization of rubber gathering.

A Provincial Study: Fordism in Michigan

As the birthplace of Fordism, however defined, industrial Michigan bears the imprint of its early development. From the very beginning, Fordism's essence as a system of stratification upheld by new norms of reproduction and consumption was clear. This section presents a critical historiography of the early years of Fordism in industrial Michigan by drawing on the work of Marxist Feminists to interrogate the prevailing wage-centered labor history. From this perspective the social policies of early automotive companies are seen as a attempt to create a stratified working class - those workers who received the benefits of the Five Dollar Day and other policies became consumers of household commodities, while those who were excluded relied on supplemental family income from service work and domestic labor. This second class provided essential unskilled labor at a low price both to automotive companies and the surrounding communities.

It is often repeated that unions in late 19th and early 20th century Michigan were weak and listless - supposedly this contributed to the rise of Fordism as a new method of labor exploitation. However, this claim is challenged by Thomas Klug, in an essay in *On the Line*. As the lumber industry declined at the close of the century, economic activity in the state shifted towards the growing cities. In the 1890s, Detroit was the stove producing capital of the world, with a significant presence of other types of metalwork that would be crucial to the development of the auto industry. Most of the organizations representing workers were craft unions, and they

admitted only skilled labor to their ranks. However, the assertion that these unions were powerless is misguided. Klug notes that “although only a few employers of iron molders formally agreed to the exclusive employment of union men, many foundries were de facto closed shops” (Klug 1989: 44). The reason for this is the Iron Molders Union’s strategy of organizing the “best molders” so that any employer wanting good labor had to go through the union.

Detroit employers were troubled by what they saw as a decrease in control over the labor process itself. As the unions had a monopoly on skilled labor, which was crucial to the operation of heavy manufacturing, they were able to exercise significant influence on the operations of firms. John J. Whirl, a member of the Employers’ Association of Detroit (EAD), said that “in some of our industries the proprietors were merely the financial agents to purchase the supplies, pay the wages, and sell the manufactured products ... As to the operations of the factory, they had no more real control than if they were in no way connected with it” (Klug 1989: 49). The attempt by employers to band together and undermine the craft unions was joined with an understanding that to truly gain control of their businesses, they would need to reduce their reliance on a small pool of skilled labor: an understanding that would animate the coming Fordist revolution.

And it was a revolution. The automobile was not invented in Michigan, but Michigan arrived early on the automotive scene, propelled by easy access to machining, raw materials, and capital: in 1896 Ransom E. Olds of Lansing and Detroit’s Charles B. King and Henry Ford demonstrated viable motor powered carriages. By the early twentieth century, there were dozens of auto companies in Michigan, but they faced significant issues. Most important was employee turnover, which was over 200% in Detroit plants, reaching a high of 382% at Ford's Highland Park plant. The costs of training new workers was estimated to be between $35 and $100 (Klug

1989: 54). At the same time, cars were expensive, and so there was a limited market for the product being produced. Henry Ford's management approach targeted both of these problems.

In 1908, Ford introduced the Model T, a mass produced car with standardized parts that required less skill to make and cost far less than competitor's cars, contributing to a rise in sales. In 1913 he introduced his version of the automated assembly line, further standardizing production and reducing costs. And in 1914, he initiated the Five Dollar Day program, a wage increase that solved the problems of employee turnover and bloated training costs. In order to compete, rival firms had to adopt and even advance Fordist innovations. In the 1920s, General Motors introduced the yearly model change to stimulate consumer demand even further.

The changes that Fordism introduced to the labor market and to the shop floor have been of particular interest to labor historians. In the early twentieth century, however, labor historians focused mainly on developments within the factories or union halls. In this literature, labor and by extension laborers are taken to be self-evidently coherent classifications: generally wage-laborers, generally men, generally working for someone else. Given the broad scope of Fordist social policies, a labor history was needed with a more expansive view of the lives of laborers. In 1971, Thomas Krueger, in "American Labor Historiography," narrates a partial break with the shared paradigm of the liberal and Marxist historiography. He is concerned with a shift away from the traditional conception of labor history, citing Hyman Berman's opinion that labor history "is more than the rise or fall or the failures and successes of organizations," and David Montgomery's call to research "the cultural and intellectual life of the working classes" (Krueger 1971: 281-2). This new direction follows the work of the British Marxists in expanding the category of the working classes to include a greater sample of the relations of working class life,

including community support of labor against capital during strikes, the influence of religion, and the permeability of class categories.

The inclusion of social questions into labor history positions the domestic sphere and its unpaid labor as a legitimate object of study. The family life of workers is not a realm separate and apart from their labor but is constitutive of it. Feminist historians have contested that the unpaid work of women and children forms the base on which the institution of waged labor stands. Joan Scott notes that "[f]eminist scholars pointed out early on that the study of women would not only add new subject matter but would also force a critical reexamination of the premises and standards of existing scholarly work" (Scott 1988: 29). The task for feminist labor history is to interrogate the very process of categorization. Elizabeth Faue, introducing a volume of essays on gender and labor history, claims that "labor historians' use of collectivities as units of analysis – whether classes, communities, trades, or families – masked both intraclass division and an understanding of conflicting interests that was at times one-dimensional" (Faue 1993: 171). To remedy this, we must examine not how the category of gender *acts on* the category of labor, but how it permeates and constitutes it.

The work of Marxists in theoretically elaborating the role of the household in the reproduction of capitalism is useful here. Marxist feminists have critically investigated the categories of the family and the household, and how they have been manipulated to serve the interests of capital. Within the household, labor is reproduced through a combination of inputs, chief among them gendered and devalued domestic labor. The structure of labor reproduction is not constant, but changes in response to different circumstances/locations. In the work of Social Reproduction Theorists, devalued domestic labor takes center stage as the condition of the

reproduction of wage laborers, and so they make a general distinction between productive labor (generally that which is done for a wage) and unproductive/reproductive labor (that which is concerned with providing for the base needs of life, on either a waged or unwaged basis). As Nancy Fraser puts it, the unwaged work of (usually) women supports the renewal of the life of the waged worker and forms “one of [the] background conditions of possibility” of capitalist accumulation (Fraser 2017: 23).

The household, however, serves more purposes than just the reproduction of wage-laborers; they are also the ultimate sites of commodity consumption. Kirstin Munro writes that "[w]orking class households are compelled to engage in household production not only to survive, but so their members can appear continually on the market as renewed labour-power in order to continually buy more commodities" (Munro 2025: 266). Because households in capitalist societies are severed from traditional means of subsistence, they are forced to resort to commodity consumption in the maintenance of their lives. This means that "the household and its production processes are implicated in the reproduction of capitalist society" (Munro 2019: 454) - households are not merely the objects of a devastating transformation effected by capitalist society. Through commodity consumption, they continually reproduce the necessary conditions for the existence of capitalism. This is not an assignment of blame, but a diagnosis of the mutually conditioning nature of the relationship between households and the capitalist society in which they are embedded.

Marxist Feminist perspectives on the household grant insight into the management of labor in Detroit around the turn of the century. Prior to 1914, when Henry Ford instituted the famous Five Dollar Day in an attempt to reduce employee turnover, it was uncommon for

Michigan families to live on the wage of a single earner. Working-class families cobbled together livings from various sources, fighting both to make higher wages and to carve out more time to do necessary subsistence labor. In "The Detroit Story: the crucible of Fordism," Kathleen Stanley and Joan Smith examine the extent to which working-class families were transformed by the family-wage in Detroit from roughly 1890 through the middle of the twentieth century (Stanley and Smith 1992). They build on the category of the household as an income-pooling unit from world-systems analysis and take the family as a site of wage-contestation. Working-class households piece together subsistence from various sources: in addition the consumption of commodities purchased with a wage, they rely on the unpaid subsistence labor of women and children, charity or community assistance, income from boarders, informal commodity production, and many other sources. The need to depend on these extra-wage sources of income would be alleviated with a family-wage.

However, after the development of the Five-Dollar Day in 1914, household patterns in Detroit changed, but not necessarily in the direction of single-income nuclear families. Stanley and Smith note that in the decades following Ford's family-wage campaign, "even in Detroit, this presumed bastion of Fordism, the family wage was never a reality for the majority of wage-earners" (Stanley and Smith 1992: 34). Except for a small elite strata of the working classes, "the male wage was always supplemented by the wages of children and sometimes of wives, as well as by considerable amounts of subsistence labor" (34). For the majority of working families, all members contributed to the maintenance of the household.

The prevalence of the family wage corresponded closely with patterns of race. Between 1900 and 1940, the population of Michigan more than doubled, from 2.4 million to 5.3 million.

During this same period, the Black population of the state increased by a factor of more than *thirteen*, from just over 15,000 to more than 208, 000 (US Census Bureau 1940). Most of this increase was driven by Black migrants from the southern states, looking for better working conditions and wages in the industrial Midwest. To an extent they were able to find them, but they did not enter the Michigan workforce on equal footing with whites. According to the 1930 census, Black men were significantly overrepresented in low paid manufacturing jobs, while their white counterparts were more often employed either as shop foremen or skilled machinists (US Census Bureau 1930). The cities themselves mirrored this, with white workers more often owning free standing homes in newly developed neighborhoods, and Black workers and poor whites confined to the slums. In other words, the social significance of Fordism is defined as much by who it excluded from prosperity as by who it included.

Global Complications: the Rubber Industry

Fordism did not develop in isolation, nor only from local dynamics. For early automobile producers in Michigan, a significant portion of the commodity chain was entirely internal to the US. Michigan's vast lumber and mineral resources were brought to Detroit by a robust rail system, and Ford even owned "mines in Kentucky and West Virginia, which gave him control over every natural resource needed to make a car - save rubber" (Grandin 2013: 118). To acquire the rubber necessary for the production of automobiles, Ford and his competitors had to look outside the US - at first to the Amazon and the Congo, and then later to Malaya and Sumatra in Southeast Asia. Initial production consisted of wild-collection by contract laborers who made long treks through the jungle to collect rubber from sparsely distributed trees. Rubber plants were quickly exported to tropical regions around the globe such as southeast Asia, although there

was little commercial production in Asia until the second decade of the twentieth century (Coates 1988: 155). Although plantations would dominate the global rubber production by the second decade of the twentieth century, for the first several decades of the auto industry there were few rationalized rubber plantations, and rubber was sourced from wild collectors in the Amazon and the Congo. By the end of the nineteenth century, the Amazon was supplying the world market with 15,000 tons of rubber annually. This section incorporates the rubber workers of the tropics into the history of Fordism by looking at the parallel dynamics of social reproduction that emerged in these regions as the growing automotive industry drove increased rubber production.

Just as Fordist production methods were dependent on racial and gendered divisions in the local labor market, they were also built on what Wallerstein refers to as the "axial division of labor," whereby the more brutal exploitation of workers in peripheral states propped up the productive relations in the core (Wallerstein 2004: 60). Although the automotive industry was made not only possible but profitable through the exploitation of rubber workers in the Global South, these workers were not included in the prosperity of Fordism. The nature of this global dynamic between the capitalist core and periphery was characterized by Andre Gunder Frank as the development of underdevelopment. In his account, the supposed economic backwardness of the Global South is not a failure to keep pace with the capitalist core, but rather “is the result of its centuries-long participation in the process of world capitalist development” (Frank 1966: 21). In other words, the underdevelopment of the Global South is a condition positively created by the dynamics of global capitalism. Samir Amin’s *Accumulation on a World Scale* deals with this concept as well, insisting that it is the capitalist core’s economic growth itself that creates the impoverished condition of the periphery. He claims that “as economic growth proceeds, none of

these features by which the structure of the periphery is distinguished lessens; on the contrary, each increases…. In the periphery growth is *not* development, for its effect is to disarticulate" (Amin 1971: 18).

This dynamic can be traced along commodity chains: global networks of labor and raw-material inputs that feed into the production of particular industrial and consumer goods. The term commodity chain was coined in a 1977 paper by Wallerstein and Hopkins, who propose it as an analytical tool to combat a "developmentalist" perspective in which "the world is said to consist of a number of related but basically autonomous 'societies' ... each moving upward along an essentially similar path of development" (Hopkins and Wallerstein 1977: 111). Criticizing an analytical nationalism which attempts to abstract autonomous societies from each other, Wallerstein and Hopkins insist on analyzing the world-system *in toto*: "a definite 'world', a spatio-temporal whole, whose spatial scope is coextensive with the elementary division of labor among its constituent regions or parts and whose temporal scope extends for as long as the elementary division of labor continually reproduces the 'world' as a social whole" (112). Within this global social body, commodity chains form a sort of vascular system, transporting raw materials and ossified labor inputs from their sites of production, combining them, and bringing them to market where they can be valorized and turned into profit.

The working conditions in rubber fields of the Amazon were a world away from those in the US. In the late 1870s a severe drought struck Northeastern Brazil, destroying the means of subsistence for a large portion of the population. Many left their homes and found work in the growing Brazilian rubber industry, but under the harshest of conditions. Forced to pay for their own provisions on the way to the work locations, rubber tappers started working in severe debt.

Further, the workers were cut off from their previous communities, and so to survive they were forced to rely on local rubber agents. Their debt was perpetuated by the sale of "the necessary working tools and basic goods such as foodstuff, which were forcibly acquired at the rubber patron's storehouse or station" (Cardoso de Mello and Melkebeke 2019: 148). The rubber companies held this debt on account against future rubber yields, but "every time [the worker's] rubber (his credit) was collected, it was found that his debt had mounted higher than his earnings" (Coates 1988: 94). The rubber agents and traders ensured that the prices of necessary provisions were higher than the piece-wages for rubber, creating a vicious cycle of debt peonage. It was this debt relation, rather than a wage, which ensured the continual supply of labor for the rubber industry.

The labor supply in the Congo was ensured through similar means, but this time with a tax burden rather than a debt burden. In a comparative study of Amazonian and Congolese rubber extraction, de Mello and Van Melkebeke note that from the 1890s on, an "in kind rubber tax … proved to be the solution to increase revenue." They go on to say that "this huge increase in exports received by in-kind taxation would result in severe and violent labour mobilisations and would cause devastation throughout the Congolese countryside" (Cardoso de Mello and Melkebeke 2019: 143). Whereas in the Amazon, workers migrated to the rubber regions due to harsh economic and environmental conditions back home, in the Congo the state and private firms sent expeditions into the countryside to capture and enslave Indigenous men. "Expeditions often ended up in plunder, destruction of villages and capturing of people: initially, to collect ivory (by the Congo Free State) and subsequently, to collect rubber (by the Congo Free State and private sector)" (147). These workers were then forced to collect and deliver "six to eight

kilograms per person every two weeks" and "when they did receive some sort of remuneration, this was usually in-kind (salt, knives, beads, textiles, etc.)" (152-53).

Unlike in the US Midwest, few provisions were made for generational reproduction. In the Amazon and the Congo, existing communities and their social relations were violently disrupted by rubber barons and state actors who saw no need to recreate conditions for the renewal of laborers. They preferred to exploit indigenous and migrant populations as an external pool from which new labor could be drawn. The pay given to rubber workers (if it can be called pay) was insufficient to provide for the renewal of the conditions of labor on an open labor market, and so the gap was filled by coercion, and by the workers' informal reproductive labor. Furthermore, although workers along the commodity chain were increasingly commodity consumers, unlike workers in Michigan who purchased household appliances with an earned wage, rubber workers obtained low quality foodstuffs and tools from closed company shops (Cardoso de Mello and Melkebeke 2019: 153).

The labor practices in wild rubber fields led to the eradication of the laboring populations. De Mello and Melkebeke seem to indicate that the 19th century practices of slaving raids in the Congo, in which "people were captured and enslaved, usually men, but also women, who were incorporated into the raiders' harems," were continued by rubber producers in the 20th century (Cardoso de Mello and Melkebeke 2019: 145-46). However, women also occasionally worked in the rubber fields. As a result of this system, the Congo witnessed a significant demographic decline, with "contemporary observers and present-day scholars [agreeing] on rural depopulation and decimation of the Congolese," along with "numerous cases of depopulated villages" (160). The situation in the Amazon was analogous: "In general terms, the demographic

post-boom scenario is marked by the emptying of villages or whole regions, being caused either by workforce recruitment, the resulting unsustainable livelihood, death by diseases, ethnocide, demographic replacement and miscegenation or encroachment" (161).

As wild harvested rubber declined on the world stage, hampered by the South American blight in the Amazon, King Leopold's personal finances in the Congo, and the low productivity of wild rubber collection, it was replaced by plantation rubber in southeast Asia. Rubber plants were first brought to Malaya in the 1870s, but were not cultivated on a large scale until 1897-8 when plantation owners were looking for a replacement for failing coffee crops (Drabble 1967: 54). With the Malay Reservations Enactment in 1913, foreigners were allowed to purchase land in certain areas and large rubber plantations developed, requiring a large immigrant workforce. To supply this demand, the government of the Federated Malay States proposed a policy centered around three points: the development of a free workforce; the recruitment of immigrant workers from India, and regulation of employment conditions, e.g. hospitals, water, housing etc. (59).

In practice, this policy led to significantly better working conditions than those in the Amazon and the Congo, but worse conditions than those of auto workers in Michigan. In the first half of the 20th century, millions of low-caste Tamils crossed the Bay of Bengal looking for work on the rubber plantations in Malaya (Amrith 2013: 2). Their legal status fluctuated between indentured, debt-bonded, and free, but these labels often had little influence over their day-to-day working conditions. J. Norman Parker, in *Colonial Labor Policy*, notes that in Malaya, "to be a free laborer often had little real meaning since the working conditions of a free laborer and an indentured laborer differed little or not at all" (Parmer 1960: 16). Most of the workers,

indentured or free, worked for a wage that was differentiated on type of job and the sex and race of the worker. Women generally received a fraction of their male co-workers wages, and Chinese laborers received more than Indian or Javanese workers. For all workers, the workday was limited, although planters struggled to extend it.

In Malaya, a steady supply of workers was ensured by the enticement of good pay, and "in the possibility of his being able, during his two or three years average stay in this country, to remit small amounts of money to relations" (Parmer 1960: 177). An interesting feature of the Malayan rubber industry is the recruiter/overseer called a *kangany* (Roy 2008: 971-98). These individuals would travel back to India to recruit family members as rubber workers, and would later serve as their managers on the plantations, employing a combination of coercions based on debt and familial relations. As in the Congo and the Amazon, the labor force was not reproduced locally and generationally, but through tapping an external source of potential workers. Unlike in the wild cultivation fields across the world, workers in Malaya were paid in cash, and employers were directed to provide for workers' housing, sanitation and medical care in addition to wages (Parmer 1960: 177-8).

The question of social reproduction is complicated by the introduction of migrant and indentured labor. As with seasonal/transhumant labor in the Amazon, the social reproductive dynamics of migrant workers in colonial Malaya were stretched across vast distances. The existing communities of workers in India and Malaya were disrupted by migration, but do not seem to have been destroyed as in the Congo and among indigenous Amazonians. Rather, the social reproduction of the workforce became bifurcated, operating locally within the amenities of the rubber plantations and regionally through cash transfers across borders. This introduces

significant complications to the question of labor reproduction, but provides an exciting opportunity to examine the fundamental relationship between labor and capital as it is attenuated by geographical and political space. By bringing the diversity of patterns of social reproduction in the rubber industry into conversation with the more traditionally understood Fordist social reproduction regime in industrial Michigan, we can see the fundamental role that labor stratification plays in the reproduction of capitalism as a system.

Conclusion

The concept of Fordism is useful in that it highlights several fundamental dilemmas facing capital at the turn of the twentieth century. The expansion of the production of commodified domestic goods was combined with an expansion in the market for their consumption. However, although the networks of consumption and production were global, the techniques used to effect their expansion were not uniform across the globe. As we see in the early automotive industry, capital employed several distinct methods of exploitation that corresponded to different cultural, political, and ecological conditions. Recent theoretical work on the household allows us to understand the role that diverse structures of reproduction play in this global network of production.

The primary question posed by this article is that of the articulation between capital's need for specific kinds of labor, and the specific ways in which it reproduces the corresponding different kinds of laborer. How did global configurations of social reproduction respond to the Fordist revolution in production? Following from the insights of Marxist Feminists, I posit that capital made use of different household patterns to structure the reproduction of labor, but this is not a simple dynamic to investigate. It is clear that neither conditions of production nor

reproduction were exported globally wholesale by Fordist producers, and yet the entire world of production was certainly transformed.

In the Amazon and the Congo, the reproduction of the debt relation and of the tax relation is inextricable from the reproduction of labor. Although commodity consumption was introduced through the closed provision shops, household production was destabilized, rather than supported. The razing of homes by slaving expeditions in the Congo, and the "kidnapping of indigenous infants … to be raised within the families of enforcers or patrons to work the rubber fields" (Cardoso de Mello and Melkebeke 2019: 151) in the Amazon are the logical parallels of the planned communities in Michigan. In Michigan a labor supply was ensured and renewed through the new family patterns of the middle class; in the Global South this reproduction of labor was ensured and renewed through the destruction of indigenous families.

The supply of labor on the Malayan plantations was also secured from external sources, but through economic enticement rather than brute force. Plantation workers were paid a wage, but as their housing was provided for by the planters, and the state enforced labor laws regarding the length of the workday and workweek. The tenure of migrant workers was short, and for many workers a portion of his/her wage was used to support families back home in India.

These patterns of reproduction are connected across the world-system by commodity chains. A finished car needed inputs of iron, wood, coal, and rubber, and each of these had to be gathered and prepared by laborers far from auto factories. The labor processes, and the conditions of its renewal, were conditioned by various local factors - the position in the global economy, the extant patterns of life and labor, and the family formation in which laborers lived. While factory workers in America could be said to be wage-earning proletarians, slave laborers

and informal workers in the rubber fields could not. And while production for the world market dominated the lives of workers and their families, it was not the only type of work they engaged in: they were supported by unique combinations of wage labor, unpaid reproductive work, and subsistence labor. If Fordism can be seen as a reconfiguration of the articulation between patterns of consumption and production, then the changes that we have traced across the Global South form a part of the history of Fordism and warrant further study within this framework.

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